

A refined method for the meta-analysis of controlled clinical trials with binary outcome[‡]

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SUMMARY

For the meta-analysis of controlled clinical trials with binary outcome a test statistic for testing an overall treatment effect is proposed, which is based on a refined estimator for the variance of the treatment effect estimator usually used in the random-effects model of meta-analysis. In simulation studies it is shown that the proposed test keeps the prescribed significance level much better than the commonly used tests in the fixed-effects and random-effects model, respectively. Moreover, when using the test it is not necessary to choose between fixed effects and random effects approaches in advance. The proposed method applies in the same way to the analysis of a controlled multi-centre study with binary outcome, including a possible interaction between drugs and centres. Copyright © 2001 John Wiley & Sons, Ltd.

1. INTRODUCTION

In this paper we consider the meta-analysis of controlled clinical trials with two treatments, say verum and placebo, where the outcome measure is a function of the number of successes in each treatment group. Possible measures are, for example, the risk difference, the (logarithmic) relative risk, or the (logarithmic) odds ratio. The methods considered apply in the same way to the analysis of a controlled multi-centre study with an outcome of this kind.

Let $\hat{\theta}_i$, $i = 1, \dots, k$, denote the estimated outcome measure in the i th study or centre; then we assume that at least approximately it holds

$$\hat{\theta}_i \sim N(\theta, \tau^2 + \sigma_i^2), \quad i = 1, \dots, k \quad (1)$$

where θ is the true underlying mean, which is generally the parameter of interest, $\tau^2 \geq 0$ is the true between-study variance and $\sigma_i^2 > 0$ is the true within-study variance. If $\tau^2 = 0$ then

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model (1) is often called the fixed effects model [1] (FE model), and if $\tau^2 > 0$, the random effects model [1] (RE model). The parameter τ^2 is also named as the variance of a random interaction of response by centres or studies [2].

Furthermore, let $\hat{\sigma}_i^2$, $i = 1, \dots, k$, denote the estimated within-study variance of the estimated outcome measure $\hat{\theta}_i$ in the i th study. Either $\hat{\sigma}_i^2$ is known or can be calculated, if, for instance in studies with binary outcome, the number of patients and successes are known.

The purpose of this paper is to suggest a test statistic for the test on the overall treatment effect θ , that is, $H_0: \theta = 0$, in controlled clinical trials with binary outcome, which satisfactorily keeps the prescribed significance level irrespective of the true between-study variance τ^2 , that is, a choice between the FE model and the RE model, in advance, is not necessary. The proposed test statistic is based on the weighted least-squares estimator of the overall treatment effect and on a refined estimator of the variance of this estimator.

There exists an extensive literature about the estimation of the between-study variance τ^2 . Throughout this paper we will work with the DerSimonian and Laird method of moments estimator [1], based on the test statistic of homogeneity originally proposed by Cochran [2], due to its ease of computation and our experience that the presented results are nearly invariant with respect to the choice of the between-study variance estimator. Other methods for estimating τ^2 are based on likelihood or Bayesian approaches [3–7]. Furthermore, MINQUE-type estimators [8–10] for τ^2 can be deduced where among these also a non-negative minimum biased quadratic estimator is derived [10, 11].

The outline of the paper is as follows. In the next section we present a real data example of placebo-controlled clinical trials of a gastroprokinetic agent in the treatment of non-ulcer dyspepsia. Section 3 contains the description of the commonly used test statistics in the FE and the RE model as well as the proposal of the new test statistic in a general framework. In Section 4 the explicit estimation formulae for the difference of probabilities, the log relative risk and the log odds ratio are given, which we also use in the simulation study. The results of the simulation study are put together in Section 5. Then, the results of the meta-analysis of the real data example are presented. Finally, we give some concluding remarks.

2. CLINICAL TRIALS ON NON-ULCER DYSPEPSIA

Non-ulcer dyspepsia is characterized by a variety of upper abdominal symptoms in the absence of an organic disease. Within the general population, dyspepsia is very common [12] and as a result empirical therapy without prior diagnostic procedures has been recommended for the management of patients with dyspepsia. Amongst other drugs the gastroprokinetic agent named cisapride has been suggested as first line empirical therapy. Therefore, we are interested in randomized placebo-controlled clinical trials with cisapride.

Scanning of the literature yields 13 placebo-controlled trials with cisapride which are eligible for a meta-analytical combination due to, for instance, definition of non-ulcer dyspepsia, study design, duration of treatment and definition of success. The results of these trials are put together in Table I.

Table I. Results of 13 cisapride studies (number of successes/number of patients) eligible for a numerical meta-analytical combination.

Study	Cisapride	Placebo
Creytens [13]	15/16	9/16
Milo [14]	12/16	1/16
Francois and De Nutte [15]	29/34	18/34
Deruyttere <i>et al.</i> [16]	42/56	31/56
Hannon [17]	14/22	6/22
Roesch [18]	44/54	17/55
De Nutte <i>et al.</i> [19]	14/17	7/15
Hausken and Bestad [20]	29/58	23/58
Chung [21]	10/14	3/15
Van Outryve <i>et al.</i> [22]	17/26	6/27
Al-Quorain <i>et al.</i> [23]	38/44	12/45
Kellow <i>et al.</i> [24]	19/29	22/30
Yeoh <i>et al.</i> [25]	21/38	19/38

3. TESTS ON THE OVERALL TREATMENT EFFECT

Let us denote by

$$w_i = \frac{1}{\tau^2 + \sigma_i^2}, \quad i = 1, \dots, k \quad (2)$$

the inverse of the variance of $\hat{\theta}_i$ in model (1). Then the weighted least-squares estimator of θ is given by

$$\hat{\theta} = \sum_{i=1}^k \frac{w_i}{w} \hat{\theta}_i, \quad w = \sum_{i=1}^k w_i \quad (3)$$

and $\hat{\theta}$ is at least approximately normally distributed with mean value θ and variance $1/w$. Thus, we reject the hypothesis of no treatment effect at level α , if

$$\psi = \frac{|\hat{\theta}|}{\sqrt{(1/w)}} > u_{1-\alpha/2} \quad (4)$$

where u_γ denotes the γ -quantile of the standard normal distribution.

In the FE model the weights w_i in (2) reduce to $v_i = 1/\sigma_i^2$, which are estimated by $\hat{v}_i = 1/\hat{\sigma}_i^2$, so that the feasible estimator in the FE model is

$$\hat{\theta}_{\text{FE}} = \sum_{i=1}^k \frac{\hat{v}_i}{\hat{v}} \hat{\theta}_i, \quad \hat{v} = \sum_{i=1}^k \hat{v}_i \quad (5)$$

and the hypothesis of no treatment effect is rejected if

$$\psi_1 = \frac{|\hat{\theta}_{\text{FE}}|}{\sqrt{(1/\hat{v})}} > u_{1-\alpha/2} \quad (6)$$

In the RE model the between-study variance has to be estimated in addition. The method of moments estimator [1] for τ^2 is given in its truncated version by

$$\hat{\tau}^2 = \max \left\{ 0, \frac{\sum_{i=1}^k \hat{v}_i (\hat{\theta}_i - \hat{\theta}_{FE})^2 - (k-1)}{\hat{v} - \sum_{i=1}^k \hat{v}_i^2 / \hat{v}} \right\} \quad (7)$$

Replace (7) and $\hat{\sigma}_i^2$ in (2) yields the feasible estimator in the RE model

$$\hat{\theta}_{RE} = \sum_{i=1}^k \frac{\hat{w}_i}{\hat{w}} \hat{\theta}_i, \quad \hat{w} = \sum_{i=1}^k \hat{w}_i, \quad \hat{w}_i = (\hat{\tau}^2 + \hat{\sigma}_i^2)^{-1}, \quad i = 1, \dots, k \quad (8)$$

so that the hypothesis of no treatment effect is rejected if

$$\psi_2 = \frac{|\hat{\theta}_{RE}|}{\sqrt{(1/\hat{w})}} > u_{1-\alpha/2} \quad (9)$$

In the just described commonly used tests, the variance of the weighted least-squares estimator $\hat{\theta}$ is estimated by estimating its components separately and this estimator is used as estimate of the variance of $\hat{\theta}_{FE}$ and of $\hat{\theta}_{RE}$, respectively, in the test statistics ψ_1 and ψ_2 depending on the model under consideration. Thus, this variance estimate does not reflect that the weights in $\hat{\theta}_{FE}$ and $\hat{\theta}_{RE}$ have been estimated; especially in the RE model, the uncertainty of the estimator of the heterogeneity parameter has not been taken into consideration.

Therefore, we first consider an estimator for the variance of $\hat{\theta}$ in model (1) following the theory of variance components estimation [11, 26]. An estimator of the variance $1/w$ of $\hat{\theta}$ in model (1) which is non-negative, unbiased, invariant with respect to mean value translations and quadratic in the study-specific estimators $\hat{\theta}_i$, $i = 1, \dots, k$, is given by

$$q = \frac{1}{k-1} \sum_{i=1}^k \frac{w_i}{w} (\hat{\theta}_i - \hat{\theta})^2 \quad (10)$$

Note that this estimator can be considered as a weighted extension of the usual empirical variance of the study-specific estimators $\hat{\theta}_i$, $i = 1, \dots, k$.

In Hartung [27] it is shown that assuming normality of $\hat{\theta}_i$, $i = 1, \dots, k$, the quadratic form $w(k-1)q$ is χ^2 -distributed with $k-1$ degrees of freedom and stochastically independent of $\hat{\theta}$, so that

$$(\hat{\theta} - \theta) / \sqrt{q} \quad (11)$$

is t -distributed with $k-1$ degrees of freedom.

By replacing the unknown variance parameters in (11) through appropriate estimates, we propose the following test. Reject the hypothesis of no treatment effect if

$$\psi_3 = \frac{|\hat{\theta}_{RE}|}{\sqrt{\hat{q}}} > t_{k-1; 1-\alpha/2} \quad (12)$$

where $\hat{q}$ denotes the feasible estimator of q with estimated weights $\hat{w}_i$, $i = 1, \dots, k$, according to (8) and $t_{v;\gamma}$ stands for the γ -quantile of the t -distribution with v degrees of freedom.

Note that the variance estimate $\hat{q}$ is zero, if all trials yield identical estimates of the treatment effect, and so the test ψ_3 in (12) is not defined. This may occur especially for binomial data

if the number of trials is rather small and the true probabilities of success are near zero or one. If $\hat{q}=0$, we extend the decision rule in (12) that in this situation we use the test ψ_1 and the corresponding decision rule from (6).

Furthermore, note that the commonly used tests ψ_1 and ψ_2 , respectively, as well as the new proposed test ψ_3 , are all approximate tests, especially for binary data, and so, in the simulation study in Section 5, the tests are compared using their actual significance level.

4. OUTCOME MEASURES FOR BINARY DATA

In this section we present the estimation formulae of the three different outcome measures for binary data, which we also use in the simulation study. The outcome measures are the risk difference, the logarithmic relative risk and the logarithmic odds ratio [1, 28].

Let us consider

$$m_{ij} \sim B(n_{ij}, p_{ij}), \quad i = 1, \dots, k; \quad j = 1, 2 \quad (13)$$

that is, the number of successes m_{ij} in treatment group j in the i th study is binomially distributed with parameters equal to the number of patients n_{ij} and the true probability of success p_{ij} in the treatment group j of the i th study.

4.1. Risk difference

The estimate of the probability of success in each treatment group is given by $\hat{p}_{ij} = m_{ij}/n_{ij}$, $i = 1, \dots, k$, $j = 1, 2$, with variance $\text{var}(\hat{p}_{ij}) = p_{ij}(1 - p_{ij})/n_{ij}$. Replacing the estimate $\hat{p}_{ij}$ in $\text{var}(\hat{p}_{ij})$ would yield an estimate of zero if $m_{ij} = 0$ or $m_{ij} = n_{ij}$. To overcome this problem we add small positive constants to the numerator and denominator of $\hat{p}_{ij}$ and work with the following estimator of the variance of $\hat{p}_{ij}$:

$$\widehat{\text{var}}(\hat{p}_{ij}) = \frac{1}{n_{ij}} \left(\frac{m_{ij} + 1/16}{n_{ij} + 1/8} \right) \left(\frac{(n_{ij} - m_{ij}) + 1/16}{n_{ij} + 1/8} \right) \quad (14)$$

which is always positive and ensures the symmetric behaviour of the variance about $p = 0.5$. Thus, we get in each study as an estimated outcome measure

$$\widehat{\text{PD}}_i = \hat{p}_{i1} - \hat{p}_{i2} = \frac{m_{i1}}{n_{i1}} - \frac{m_{i2}}{n_{i2}}, \quad i = 1, \dots, k \quad (15)$$

with estimated variance of

$$\widehat{\text{var}}(\widehat{\text{PD}}_i) = \widehat{\text{var}}(\hat{p}_{i1}) + \widehat{\text{var}}(\hat{p}_{i2}) \quad (16)$$

using formula (14) for each component on the right hand side of (16).

4.2. Logarithmic relative risk

Following Pettigrew *et al.* [29] we estimate the log relative risk of two probabilities of success in each study with

$$\log \widehat{\text{RR}}_i = \log \frac{(m_{i1} + 0.5)(n_{i2} + 0.5)}{(n_{i1} + 0.5)(m_{i2} + 0.5)}, \quad i = 1, \dots, k \quad (17)$$

and the estimated variance of this estimator is

$$\widehat{\text{var}}(\log \widehat{\text{RR}}_i) = \frac{1}{m_{i1} + 0.5} - \frac{1}{n_{i1} + 0.5} + \frac{1}{m_{i2} + 0.5} - \frac{1}{n_{i2} + 0.5} \quad (18)$$

The variance estimator is always positive if $m_{i1} \neq n_{i1}$ or $m_{i2} \neq n_{i2}$. If $m_{i1} = n_{i1}$ and $m_{i2} = n_{i2}$, then in (18) we do not add 0.5 to m_{ij} , $j = 1, 2$, in accordance with Pettigrew *et al.* [15] to ensure the positiveness of the variance estimator.

4.3. Logarithmic odds ratio

The estimator of the log odds ratio following Gart and Zweifel [30] is given by

$$\log \widehat{\text{OR}}_i = \log \left(\frac{m_{i1} + 0.5}{(n_{i1} - m_{i1}) + 0.5} \frac{(n_{i2} - m_{i2}) + 0.5}{m_{i2} + 0.5} \right), \quad i = 1, \dots, k \quad (19)$$

with an estimated variance of

$$\widehat{\text{var}}(\log \widehat{\text{OR}}_i) = \frac{1}{m_{i1} + 0.5} + \frac{1}{(n_{i1} - m_{i1}) + 0.5} + \frac{1}{m_{i2} + 0.5} + \frac{1}{(n_{i2} - m_{i2}) + 0.5} \quad (20)$$

5. SIMULATION STUDY

In a simulation study we compare the three test statistics ψ_1, ψ_2, ψ_3 concerning their actual significance levels given a prescribed significance level of $\alpha = 0.05$ for the three different outcome measures considered in the previous section. We focus on the meta-analytical combination of either $k = 6$ or $k = 12$ trials. The simulation study was carried out in SAS 6.12 using the random number functions RANNOR and RANBIN. Each reported estimated significance level is based on 10 000 replications of the corresponding model.

The data in each study are generated with the following procedure. Let us denote by p_1 and p_2 the true overall probabilities of success in the verum and placebo group, respectively. The parameter of interest θ is given as the difference of $f(p_1)$ and $f(p_2)$, where f is a suitable chosen function, for instance the identity, logarithmic, or logit function depending on the considered parameter of interest. The study-specific treatment effect $\theta_i = f(p_{i1}) - f(p_{i2})$, $i = 1, \dots, k$, is generated using that $f(p_{ij})$, $i = 1, \dots, k$, $j = 1, 2$, is normally distributed with mean $f(p_j)$ and variance $\tau^2/2$, where τ^2 is the parameter of heterogeneity. With the appropriate inverse function f^{-1} , we yield the study-specific probabilities p_{ij} , $i = 1, \dots, k$, $j = 1, 2$, and the number of successes in each trial and each treatment group is then independently generated from a binomial distribution.

The patterns of sample sizes used in the simulation study are put together in Table II. The first two patterns have an equal number of patients in each treatment group and in each study, that is, under the hypothesis that the overall probabilities of success p_1 and p_2 are identical, the study-specific estimators $\hat{\theta}_i$ theoretically contribute identical weights w_i/w to the meta-analysis irrespective of the value of the parameter of heterogeneity. Thus, with the second pattern we can give an impression to what extent an increasing number of patients within a treatment group influences the results of the meta-analysis. The third pattern contains some smaller trials and some larger trials so that in this situation, assuming identical probabilities p_1 and p_2 and no heterogeneity is present, the larger trials contribute a large part to the result

Table II. Patterns used in the simulation study.

Pattern	$k = 6$ trials $(n_{i1}, n_{i2}), i = 1, \dots, 6$						$k = 12$ trials
1	(20, 20)	(20, 20)	(20, 20)	(20, 20)	(20, 20)	(20, 20)	once replicated the patterns for $k = 6$ trials
2	(100, 100)	(100, 100)	(100, 100)	(100, 100)	(100, 100)	(100, 100)	
3	(20, 20)	(20, 20)	(20, 20)	(100, 100)	(200, 200)	(200, 200)	
4	(20, 20)	(40, 40)	(60, 60)	(80, 80)	(100, 100)	(120, 120)	

of the meta-analysis. The fourth pattern with different sample sizes in each study reflects a moderate unbalanced structure of the weights which each study contributes to the meta-analysis given $p_1 = p_2$ and no heterogeneity.

Owing to the fact that the three outcome measures have different measurement scales, the chosen values of the parameter of heterogeneity are given in the corresponding subsections.

5.1. Results for the risk difference

First, we present the results of our simulation study for the probability difference approach. An extract of the results is shown in Table III. In this table we confine ourselves to the true probabilities $p_1 = p_2 = p = 0.1$ and 0.5 , which reflects from a theoretical point of view a rather small variance and the largest possible variance of the outcome measure, respectively. Note that the variance of the risk difference is symmetric about $p = 0.5$ so that an interchange of the definitions of *success* and *no success* would just change the sign of the estimate of the outcome measure but the variance estimate remains the same.

Considering the simulation scheme for the probability difference approach, we have to account for the fact that the study-specific probabilities p_{ij} should lie between zero and one. Thus, conditional on the chosen p we pick out the values of the heterogeneity parameter τ^2 and truncate the simulated study-specific probabilities at 0.001 and 0.999 , respectively. For $p = 0.1$ we consider the values $\tau^2 = 0, 0.001$ and 0.01 and for $p = 0.5$ the values $\tau^2 = 0, 0.01$ and 0.05 .

Let us first consider the results in Table III if the heterogeneity parameter τ^2 is equal to zero, that is, the FE model is the theoretically correct one. The test ψ_1 , which is commonly used one in the FE model, has estimated sizes which are almost always larger than the prescribed significance level. In pattern 1 we observe estimated sizes up to nearly 7.5 per cent for $p = 0.5$. This phenomenon has already been observed by comparing normally distributed means with unequal variances [31, 32]. However, for increasing sample sizes the estimated sizes tend to the given level, that means, for patterns 2, 3 and 4, the test ψ_1 yields satisfactory results if the true probability of success in each treatment group is around 0.5. In the case $p = 0.1$ the test ψ_1 has estimated sizes between 5.3 per cent and 6.5 per cent but we cannot state a clear tendency towards the prescribed level. The doubling of the number of trials does not essentially influence the attitude of the test ψ_1 .

The test ψ_2 , which is the commonly used one in the RE model, is in nearly all cases rather conservative if no heterogeneity is present. In the simulation study we mainly observe estimated sizes between 3.6 per cent and 4.5 per cent. Only for $p = 0.5$ in pattern 1 are the estimated sizes around the given level.

Table III. Estimated sizes (in per cent) of the tests ψ_1 from (6), ψ_2 from (9), and ψ_3 from (12) for $k=6$ and $k=12$ trials with the outcome measure *risk difference* given a nominal significance level of $\alpha=0.05$.

Pattern	p	τ^2	$k=6$ trials			$k=12$ trials		
			ψ_1	ψ_2	ψ_3	ψ_1	ψ_2	ψ_3
1	0.1	0	5.60	4.57	5.83	5.29	4.26	4.91
		0.001	6.12	4.91	5.40	5.21	4.12	4.20
		0.01	10.03	6.36	3.96	6.73	5.57	3.07
	0.5	0	7.07	4.95	5.21	7.37	5.22	5.54
		0.01	13.24	7.01	5.35	13.73	6.34	5.34
		0.05	36.70	10.33	5.95	38.73	7.79	5.76
2	0.1	0	6.31	4.49	5.32	6.54	4.86	5.74
		0.001	13.18	7.92	5.51	12.67	6.66	5.37
		0.01	39.70	10.66	4.63	36.21	9.31	4.82
	0.5	0	5.39	4.00	5.10	4.95	3.72	4.44
		0.01	27.11	9.53	4.97	27.79	7.63	5.21
		0.05	62.04	10.59	5.24	62.09	6.75	4.95
3	0.1	0	5.60	3.76	4.73	5.25	3.60	4.20
		0.001	16.13	8.50	6.94	15.47	7.53	6.24
		0.01	50.13	12.08	6.21	47.51	9.38	5.54
	0.5	0	5.60	3.74	4.89	5.17	3.62	4.46
		0.01	35.01	12.12	7.58	35.81	9.04	6.71
		0.05	66.99	12.68	6.60	69.21	8.20	5.71
4	0.1	0	6.49	4.54	5.26	6.37	4.59	5.17
		0.001	12.03	7.19	5.63	11.38	6.29	5.44
		0.01	35.87	10.65	5.41	33.02	9.09	4.83
	0.5	0	5.16	3.72	4.82	5.36	3.99	4.74
		0.01	26.61	10.13	5.93	26.29	7.97	5.94
		0.05	59.49	11.22	5.84	59.26	7.69	5.53

The new test ψ_3 often has estimated sizes around 5 per cent if the FE model is the correct one. Only in pattern 3 for $k=12$ trials does the test seem to be a bit conservative.

If heterogeneity is present, we give the results for the test ψ_1 just for demonstrating the known fact that in this case the test ψ_1 has a high increasing actual size with growing heterogeneity parameter τ^2 , and the use of this test is inappropriate in this situation. Thus, in the following, we confine ourselves to the comparison of the tests ψ_2 and ψ_3 .

The test ψ_2 can become very anticonservative, especially for $k=6$ trials. For instance, in pattern 3 the simulation study yields estimated sizes up to nearly 13 per cent. The doubling of the number of trials has the effect in all patterns that the estimated sizes for $k=12$ trials are lower but the test still remains anticonservative in most cases. Only in pattern 1 with $p=0.1$ are the estimated sizes smaller than 6 per cent.

The new test ψ_3 often has estimated sizes around the prescribed level. Two exceptional cases, however, have to be pointed out. In pattern 1 with $p=0.1$ the test may become conservative as indicated by the estimated size for $\tau^2=0.01$. In pattern 3, especially for $k=6$

Table IV. Estimated sizes (in per cent) of the tests ψ_1 from (6), ψ_2 from (9), and ψ_3 from (12) for $k=6$ and $k=12$ trials with the outcome measure *log relative risk* given a nominal significance level of $\alpha=0.05$.

Pattern	p	τ^2	$k=6$ trials			$k=12$ trials		
			ψ_1	ψ_2	ψ_3	ψ_1	ψ_2	ψ_3
1	0.4	0	2.20	1.98	4.23	2.52	2.33	4.04
		0.01	2.80	2.53	4.18	2.41	2.16	3.48
		0.2	5.11	4.01	4.19	4.82	3.78	3.86
	0.8	0	3.00	2.57	4.58	3.06	2.71	4.31
		0.01	3.18	2.77	4.37	3.29	2.91	4.30
		0.2	6.70	4.85	4.65	6.49	4.54	3.98
2	0.4	0	4.06	3.21	4.70	4.05	3.39	4.53
		0.01	5.02	3.86	4.56	5.26	3.93	4.33
		0.2	22.17	9.35	4.49	22.58	7.44	4.80
	0.8	0	4.65	3.58	4.51	4.49	3.51	4.84
		0.01	5.96	4.41	4.63	6.16	4.78	4.99
		0.2	26.24	9.98	5.14	25.61	7.85	4.96
3	0.4	0	4.29	3.36	4.95	4.19	3.53	5.22
		0.01	6.31	4.69	5.83	6.18	4.65	5.68
		0.2	32.17	12.54	8.59	31.61	9.31	6.81
	0.8	0	4.60	3.63	4.99	4.59	3.56	4.74
		0.01	6.69	4.80	5.53	6.76	4.73	5.40
		0.2	35.68	12.43	8.29	35.01	9.34	6.64
4	0.4	0	3.87	3.02	4.41	4.16	3.49	4.77
		0.01	4.51	3.44	4.75	4.99	3.85	4.97
		0.2	20.05	9.52	5.89	20.09	7.86	5.52
	0.8	0	4.51	3.30	4.53	4.48	3.71	5.01
		0.01	5.61	4.29	4.93	5.61	4.33	5.12
		0.2	23.85	9.61	5.40	22.98	7.42	5.07

trials, the test ψ_3 has estimated sizes between 6 per cent and 7.5 per cent and is, hence, anticonservative in this situation. However, the test ψ_3 is still superior compared to the test ψ_2 concerning the actual size for this pattern.

5.2. Results for the logarithmic relative risk

In Table IV an excerpt of the results for the logarithmic relative risk is given. As representatives for the true probabilities we choose $p=0.4$ and 0.8 in this table. Note that for increasing p the variance of the log relative risk estimate becomes smaller and in the simulation study we have observed that for the commonly used tests ψ_1 and ψ_2 the estimated sizes become larger for increasing p . As values of the heterogeneity parameter we take $\tau^2=0, 0.01$ and 0.2 .

Let us first consider the results in Table IV in the case where no heterogeneity is present. The test ψ_1 is rather conservative for the pattern 1 with largest estimated sizes around 3 per cent. For the other three patterns the test is still a little conservative with estimated actual sizes around 4 per cent for $p=0.4$ and around 4.5 per cent for $p=0.8$. The number of trials

does not essentially affect the behaviour of the test ψ_1 with regard to the actual significance level. Thus, these results indicate that for a fixed number of studies the number of patients has to be large so that the test ψ_1 is not too conservative and attains its nominal significance level.

For the commonly used test in the RE model, ψ_2 , the simulation study yields estimated sizes which are always smaller than the estimated sizes of the test ψ_1 if $\tau^2 = 0$. In all considered parameter constellations the estimated actual significance levels for the test ψ_2 do not exceed 4 per cent.

The new proposed test ψ_3 is a bit conservative in pattern 1 with estimated sizes between 4 per cent and 4.5 per cent but in all the other patterns the estimated sizes lie between 4.4 per cent and 5.2 per cent and are always closer to the prescribed level than the estimated sizes for the test ψ_1 given a fixed parameter constellation.

If heterogeneity is present, the test ψ_1 preserves its conservative attitude for moderate values of τ^2 in pattern 1 and has estimated sizes between 5 per cent and 6.7 per cent for larger values of τ^2 . For the other three patterns the simulation study shows that for moderate values of τ^2 the estimated sizes of the test ψ_1 are around the prescribed level with a slight anticonservative tendency in pattern 3. For $\tau^2 = 0.2$ the actual significance level of the test ψ_1 substantially exceeds the nominal one.

The test ψ_2 shows a conservative behaviour for moderate values of the heterogeneity parameter τ^2 , only in pattern 3 the estimated sizes are around 4.7 per cent for $\tau^2 = 0.01$. For growing τ^2 the test can become very anticonservative, except in pattern 1 where, for instance, the estimated sizes are almost always smaller than 4.6 per cent for $\tau^2 = 0.2$, but for the other three patterns the estimated sizes lie between 9.4 per cent and 12.5 per cent for $k = 6$ trials. The doubling of the number of studies has the effect that for $k = 12$ trials the estimated sizes are smaller but the test remains anticonservative.

The test ψ_3 shows a rather homogeneous attitude concerning the estimated sizes for a given pattern. For pattern 1 the test is a bit conservative, especially for $p = 0.4$, but in the other patterns the estimated sizes are in most cases around the prescribed level. Only in pattern 3 with large values of τ^2 is the test anticonservative. We have already observed this phenomenon for the risk difference, but even in this situation the estimated sizes of the test ψ_3 are smaller than the corresponding ones of the test ψ_2 .

5.3. Results for the logarithmic odds ratio

Like the variance of the risk difference estimator, the variance of the log odds ratio estimator is symmetric about $p = 0.5$ for $p_1 = p_2 = p$. Thus, in Table V we report the results for $p = 0.1$ and $p = 0.5$ as in Table III for the risk difference. As values of the heterogeneity parameter we consider $\tau^2 = 0, 0.1$ and 1.

If the heterogeneity parameter is equal to zero, the estimated sizes of the test ψ_1 lie between 1 per cent and 4 per cent for $p = 0.1$ and between 4.2 per cent and 5 per cent for $p = 0.5$. Thus, for true probabilities near zero or one the test ψ_1 can be rather conservative, whereas for $p = 0.5$ the test is only a bit conservative.

The test ψ_2 is rather conservative if no heterogeneity is present and the estimated actual significance levels in the simulation study do not exceed 4 per cent.

The best results concerning the actual significance level yield the new proposed test ψ_3 . The estimated sizes are often near the prescribed level and only for $k = 12$ trials with $p = 0.1$ does the test tends to be a bit conservative except in pattern 3.

Table V. Estimated sizes (in per cent) of the tests ψ_1 from (6), ψ_2 from (9), and ψ_3 from (12) for $k=6$ and $k=12$ trials with the outcome measure *log odds ratio* given a nominal significance level of $\alpha=0.05$.

Pattern	p	τ^2	$k=6$ trials			$k=12$ trials		
			ψ_1	ψ_2	ψ_3	ψ_1	ψ_2	ψ_3
1	0.1	0	1.15	1.09	4.35	0.93	0.92	3.51
		0.1	1.24	1.20	3.85	1.52	1.51	3.76
		1	5.87	4.58	4.72	6.24	4.83	4.48
	0.5	0	4.30	3.47	4.62	4.20	3.56	4.75
		0.1	6.01	4.40	4.58	6.09	4.40	4.39
		1	20.65	8.80	4.08	19.83	7.50	4.85
2	0.1	0	3.34	2.76	4.26	3.36	2.84	4.21
		0.1	7.31	5.15	4.45	6.93	4.51	3.96
		1	33.96	9.84	4.33	33.29	7.82	4.54
	0.5	0	4.69	3.57	4.93	4.37	3.47	4.25
		0.1	17.75	8.19	4.89	18.05	7.31	5.13
		1	53.65	10.97	4.67	53.93	8.63	5.33
3	0.1	0	3.48	2.85	5.01	4.13	3.54	5.76
		0.1	10.83	7.14	7.23	10.94	7.05	7.23
		1	44.36	14.12	8.51	44.49	9.12	6.86
	0.5	0	4.83	3.48	5.07	4.97	3.78	4.92
		0.1	24.80	10.98	7.98	25.01	8.82	6.88
		1	61.88	12.57	6.08	61.77	8.41	5.33
4	0.1	0	3.08	2.59	4.62	2.98	2.57	4.15
		0.1	6.38	4.72	5.01	6.29	4.66	4.67
		1	30.78	10.19	5.35	30.31	7.92	4.98
	0.5	0	4.49	3.44	4.73	4.53	3.58	4.69
		0.1	16.79	8.50	5.78	16.12	7.29	5.55
		1	50.11	11.55	5.52	49.95	8.55	5.12

If $\tau^2 > 0$, the test ψ_1 substantially exceeds the nominal significance level for large values of τ^2 , except in pattern 1 for $p=0.1$. For moderate values of τ^2 , like $\tau^2=0.1$, the behaviour of the test ψ_1 concerning the actual significance level is quite different. In pattern 1 the test is rather conservative for $p=0.1$, whereas in pattern 3 we observe estimated sizes up to 25 per cent. The test ψ_2 is often too anticonservative if heterogeneity is present. Above all for $k=6$ trials the estimated sizes attain values around 10 per cent. The doubling of the number of studies leads to smaller estimated sizes but the test often remains too anticonservative even for $k=12$ trials. Only for moderate values of τ^2 and $p=0.1$ in patterns 2 and 4 are the estimated sizes near the prescribed level whereas for moderate values of τ^2 in pattern 1 the test is conservative. For the new proposed test ψ_3 we obtained rather homogeneous results concerning the estimated actual significance levels within a given pattern. The estimated sizes are often around the prescribed level. In pattern 1 the test is a little conservative for moderate values of τ^2 and in pattern 3 the test shows an anticonservative attitude with, for instance, estimated sizes between 7 per cent and 8.5 per cent for $p=0.1$. However, on the

Table VI. Estimates of the three different outcome measures with estimated variances (in parentheses) in the 13 clinical trials with cisapride.

Study	Risk difference		Log relative risk		Log odds ratio	
Creytens [13]	0.3750	(0.0192)	0.4895	(0.0486)	2.0990	(0.9698)
Milo [14]	0.6875	(0.0156)	2.1203	(0.6255)	3.3570	(1.0334)
Francois and De Nutte [15]	0.3235	(0.0110)	0.4666	(0.0300)	1.5652	(0.3304)
Deruyttere <i>et al.</i> [16]	0.1964	(0.0078)	0.2995	(0.0199)	0.8640	(0.1635)
Hannon [17]	0.3636	(0.0196)	0.8023	(0.1339)	1.4656	(0.4011)
Roesch [18]	0.5057	(0.0067)	0.9515	(0.0432)	2.2326	(0.2008)
De Nutte <i>et al.</i> [19]	0.3569	(0.0252)	0.5379	(0.0806)	1.5465	(0.6057)
Hausken and Bestad [20]	0.1034	(0.0084)	0.2274	(0.0423)	0.4125	(0.1385)
Chung [21]	0.5143	(0.0254)	1.1653	(0.2475)	2.1203	(0.6832)
Van Outryve <i>et al.</i> [22]	0.4316	(0.0151)	1.0274	(0.1369)	1.8072	(0.3628)
Al-Quorain <i>et al.</i> [23]	0.5970	(0.0070)	1.1472	(0.0615)	2.7647	(0.2897)
Kellow <i>et al.</i> [24]	-0.0782	(0.0143)	-0.1098	(0.0290)	-0.3544	(0.3086)
Yeoh <i>et al.</i> [25]	0.0526	(0.0131)	0.0976	(0.0458)	0.2059	(0.2062)

whole, the test ψ_3 has estimated sizes which are closer to the prescribed level than the corresponding estimated sizes of the test ψ_2 .

6. RESULTS OF THE REAL DATA EXAMPLE

To illustrate the different test procedures from Section 3 we conduct the meta-analysis of the 13 cisapride studies, see Table I, for all three outcome measures for binary data discussed in Section 4. The study-specific estimates for the different outcome measures as well as the variance estimates given in Table VI are calculated using the formulae from Section 4 so that slight numerical differences to the published results of the outcome measure used in the original papers may occur. Only in one study [24] do the outcome measures not yield a result which is in favour of cisapride.

The results of the meta-analyses in the fixed effects approach as well as in the random effects approach are shown in Table VII, where beside the estimates of the overall treatment effects, the 95 per cent confidence intervals deduce from the three tests ψ_1 , ψ_2 and ψ_3 , and the estimates of the between-study variance τ^2 are reported. The estimates of τ^2 for all outcome measures indicate that a positive between-study variance is present, that is, it does not seem to be appropriate to assume homogeneous treatment effects over the studies. Moreover, due to the large values of the estimates of $\hat{\tau}^2$, the results of the simulation study indicate that the confidence intervals deduced from the commonly used tests may be too narrow in this example.

Irrespective of the choice of the outcome measure and the chosen method, the meta-analyses show a significant superiority of cisapride over placebo, but some aspects are worth noting. Considering the probability difference approach we observe that the estimates in the fixed effects model and the random effects model are nearly unchanged, while concerning the other two outcome measures the random effects model yields larger estimated values than the fixed effects one. For instance, the estimated odds ratio in the fixed

Table VII. Meta-analytical estimates of the overall treatment effect, corresponding 95 per cent confidence intervals deduced from the test statistics ψ_1 , ψ_2 and ψ_3 and estimated between-study variance for the 13 clinical trials with cisapride.

Model	Difference of probability	Log relative risk	Log odds ratio
<i>Fixed effects</i>			
Estimate $\hat{\theta}_{FE}$	0.34086	0.44219	1.23053
95 per cent CI (ψ_1)	(0.28143; 0.40030)	(0.31977; 0.56460)	(0.93250; 1.52856)
<i>Random effects</i>			
Estimate $\hat{\theta}_{RE}$	0.33795	0.55747	1.42088
95 per cent CI (ψ_2)	(0.21256; 0.46333)	(0.31764; 0.79729)	(0.85150; 1.99026)
95 per cent CI (ψ_3)	(0.20261; 0.47328)	(0.27286; 0.84208)	(0.79712; 2.04464)
$\hat{\tau}^2$	0.03938	0.12098	0.71755

effects approach is $\exp(1.23053) = 3.4230$, and $\exp(1.42088) = 4.1408$ in the random effects approach.

Furthermore, the estimated confidence intervals in the fixed effects models are always overlapped by the estimated confidence intervals in the random effects model despite the differences of the estimates of log relative risk and log odds ratio in both models. Moreover, the confidence intervals deduced from the test statistic ψ_3 are wider than the commonly used confidence intervals.

7. CONCLUSIONS

The proposed test shows a good performance in keeping the prescribed significance level for testing an overall treatment effect in data situations where the outcomes are binary. Although the test is originally derived under normality assumption of the outcome measure, the simulation study shows that the estimate of the variance of the estimated outcome measure as well as the approximation of the distribution of the test statistic to the distribution of the t -variate works well even for binomial distributed random variables. We do not observe a strong overestimation or underestimation of the variance depending on the parameter sets chosen, as can happen in the commonly used methods for binary outcome leading to a rather conservative or liberal attitude of the commonly used tests.

A major advantage of the proposed test is its good performance irrespective of the value of the true non-negative between-study variance. Consequently, a choice between the fixed effects and random effects approach in advance is not necessary.

Finally, it is worthwhile noting once again that we propose a new variance estimate of the overall treatment effect estimator and as the overall treatment effect estimator we use the commonly used one in the RE model. Thus, the proposed test for judging the overall treatment effect from studies with binary outcome is an improvement in that part of the meta-analytical process in which no covariates are available which may explain heterogeneity [33–35]. Furthermore, due to its easy application and its good performance in the simulation study, the proposed test is a good alternative to the tests from other approaches which also

take into account the uncertainty of the estimation of the heterogeneity parameter but require more computational effort like the likelihood or Bayesian approach [3–7].

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